

Control Regions with Beam-On Data

Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production on Argon in the NuMI Beam

First results with real data; the signal region remains blind

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Ishanee Pophale, Jarek Nowak

Lancaster University

Abstract

The four inclusive control regions of the CC $1\mu 1\pi Xp$ NuMI analysis have been opened on beam-on data. This note reports what was processed, the yields and distributions in each region, and the statistics of the frozen comparison procedure applied to the data. The signal region has *not* been opened: the beam-on files were processed into a signal-region-stripped skim, verified to contain no event passing the signal selection.

Two things must be settled before any of these numbers is read as a statement about the background model. First, **no Run-2 beam-on file exists**, so Run 2 is excluded from the data *and* from the prediction throughout; the comparison covers 85.7% of the FHC and 76.6% of the RHC exposure. Second, and more seriously, the run-by-run comparison is not uniform: **FHC Run 1 sits at a data/prediction ratio of 0.64–0.72 in the high-statistics regions while every other run period of either beam mode lies between 1.08 and 1.25**. Flux and detector uncertainties are common to the run periods of a mode and cancel in that comparison, so the difference is not a systematic of the prediction. The natural explanation is the exposure assumed for the staged FHC Run-1 beam-on file, which no quantity in this comparison verifies. Until that is resolved the mode-level ratios of §2 should not be interpreted.

With FHC Run 1 set aside, the remaining five run periods sit coherently 10–25% above the prediction in all four regions. That is a common normalisation offset, the one direction the frozen global test is explicitly insensitive to, and it is within the 20–30% prediction uncertainty. It is nonetheless coherent across regions of very different composition, and is the second thing to understand.

Applied to the data, the frozen procedure’s pass/fail statistic — the global normalisation χ^2 over the eight (region, mode) totals — gives 8.77/8, $p = 0.362$, and no region is flagged by the per-region pulls. The shape tests, which project the normalisation out and are therefore blind to all of the above, are a different matter: the muon-angle distribution fails in the CC 0π and π^0 regions in *both* beam modes, by a wide margin, while the sixteen other tests are unremarkable. The same code on pseudo-data returned all twenty-two with $p > 0.47$. Under

the frozen rules that removes both of those regions from eligibility for a fitted normalisation correction, which requires a passing shape test.

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1 What was opened, and what was not

The analysis is blind in the signal region. Beam-on data were processed with the control-region skim: an event is written only if it *fails* the complete signal-region selection *and* satisfies at least one of the four approved control-region definitions. The written file carries a marker that the cross-section extraction refuses, and the sideband tools accept a beam-on file only if it carries that marker. Each processing job re-read its own output and deleted it unless the marker was present, no written event carried the signal-region flag, and every written event fell in at least one control region. All eight files passed.

Because the skim keeps every control-region event and discards only events that are in the signal region or in no region at all, the control-region distributions below are complete: nothing that belongs in a control region has been dropped.

Table 1: Beam-on files processed. “Input events” is the number of entries in the PeLEE ntuple; the run range is read from the file and is what fixes the horn-mode assignment of the Run-4 sub-periods, which the file names do not always carry. The beam-off files of the same production have the mode in their names and the same run ranges, which is how 4a/4b are identified as RHC and 4c/4d as FHC.

Period	File	Run range	Input events
FHC Run 1	run1_beamon_beamgood	4952–6748	610 496
FHC Run 4c	beam_on_run4c	21286–22269	392 148
FHC Run 4d	beam_on_run4d_fhc	22270–23259	228 821
FHC Run 5	beam_on_run5	24376–25759	748 669
RHC Run 1	run1_RHC_beamon_beamgood	6748–6998	167 923
RHC Run 3	run3b_beamon_beamgood	14117–17566	1 104 349
RHC Run 4a	beam_on_run4a	19660–19752	43 027
RHC Run 4b	beam_on_run4b	19753–21285	835 573

The CRT-era trigger flag. The Run-4 and Run-5 beam-on files carry a software-trigger flag that is identically zero and no companion pre-CRT flag, exactly the pattern of the CRT-era beam-off samples. The branch is bound only when the companion branch is present, so these files keep the fail-open default and are not rejected. Had that not been the case, Runs 4 and 5 would have processed to zero events silently. This was verified on the beam-on files directly before processing.

No Run-2 beam-on data. The staged files cover run numbers 4952–6998, 14117–17566 and 19660–25759. Nothing lies in the range 6999–14116, which contains all of Run 2. The exposure the prediction is normalised to assigns Run 2 1.268×10^{20} POT in FHC and 2.591×10^{20} in RHC, so the missing period is 14.3% of the FHC, 23.4% of the RHC and 19.4% of the combined exposure. Run 2 is therefore dropped from the data and from the prediction — the Monte Carlo, the beam-off and the dirt — everywhere in this note. The beam-off side of the analysis stands Run 1 in for Run 2 permanently, because beam-off is a cosmic-only sample matched by run period; that substitution is not available for beam-on data, which is the measurement.

Table 2: Exposure entering this comparison, against the full exposure the analysis is normalised to.

	Runs used	POT used (10^{20})	Fraction of the mode
FHC	1, 4, 5	7.589	85.7%
RHC	1, 3, 4	8.491	76.6%

2 Yields and distributions

The prediction is the same stack as the blind studies: neutrino Monte Carlo scaled per run to that run's data exposure with the central-value weight, plus the per-run beam-off sample scaled by its gate ratio, plus the dirt overlay. Unlike the pseudo-data studies, *nothing is added to the data*: real beam-on data already contains cosmics and dirt, which here belong to the prediction alone.

Table 3: Control-region yields with beam-on data, summed over the run periods used. The prediction components are the neutrino signal contamination, the neutrino background, the beam-off and the dirt. **The mode-level ratios in the last column mix the FHC Run-1 anomaly of §3 with the rest and should not be read on their own.**

Mode	Region	Data	Prediction	signal	ν bkg	beam-off	dirt
FHC	CC0 π	10 660	11 571.0	962.3	9 442.8	1 118.1	47.8
	multi- π	109	109.6	24.3	79.6	5.7	0.1
	π^0	6 291	6 327.3	562.2	5 119.5	635.4	10.1
	cosmic	92	108.3	5.0	59.5	42.0	1.9
RHC	CC0 π	12 047	10 558.9	915.2	8 763.9	846.7	33.1
	multi- π	136	125.0	27.1	94.3	3.6	0.1
	π^0	8 092	6 639.5	594.0	5 578.8	459.7	7.0
	cosmic	120	89.5	5.0	58.0	25.2	1.3

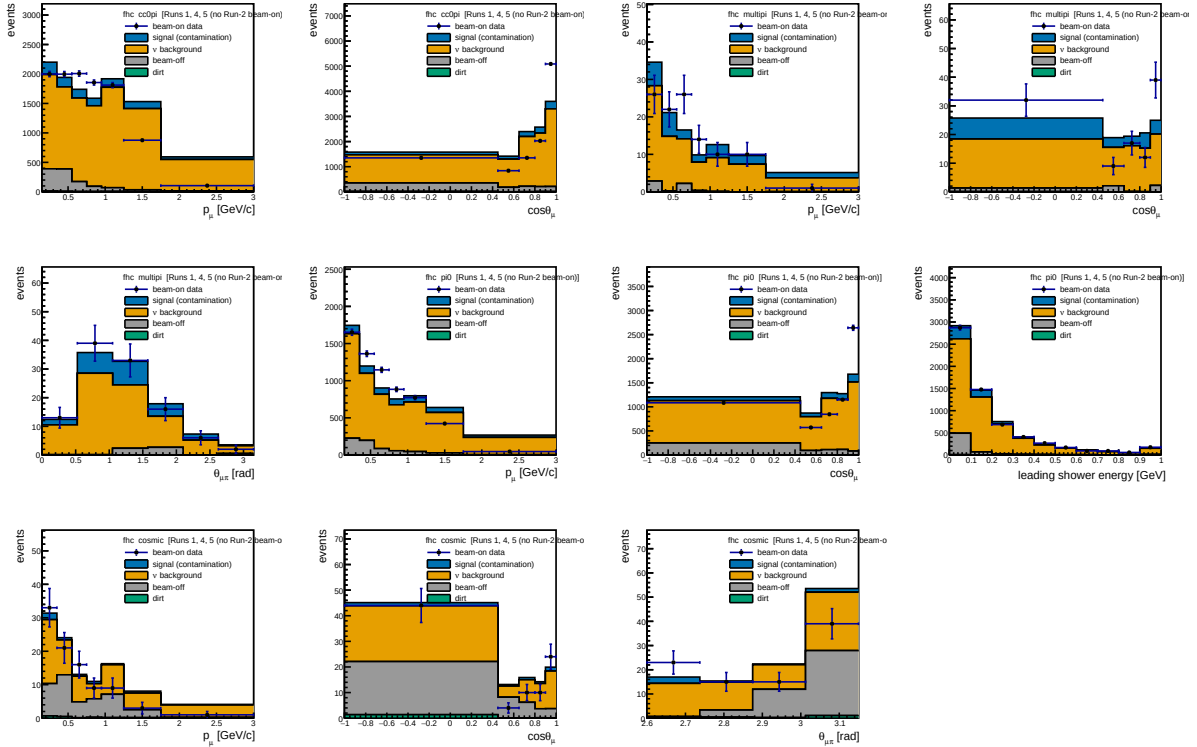


Figure 1: FHC control regions with beam-on data (Runs 1, 4, 5). Rows are the CC0 π , multi- π , π^0 and cosmic regions; the distributions are the frozen ones. Points are beam-on data with Poisson errors; the stack is the prediction.

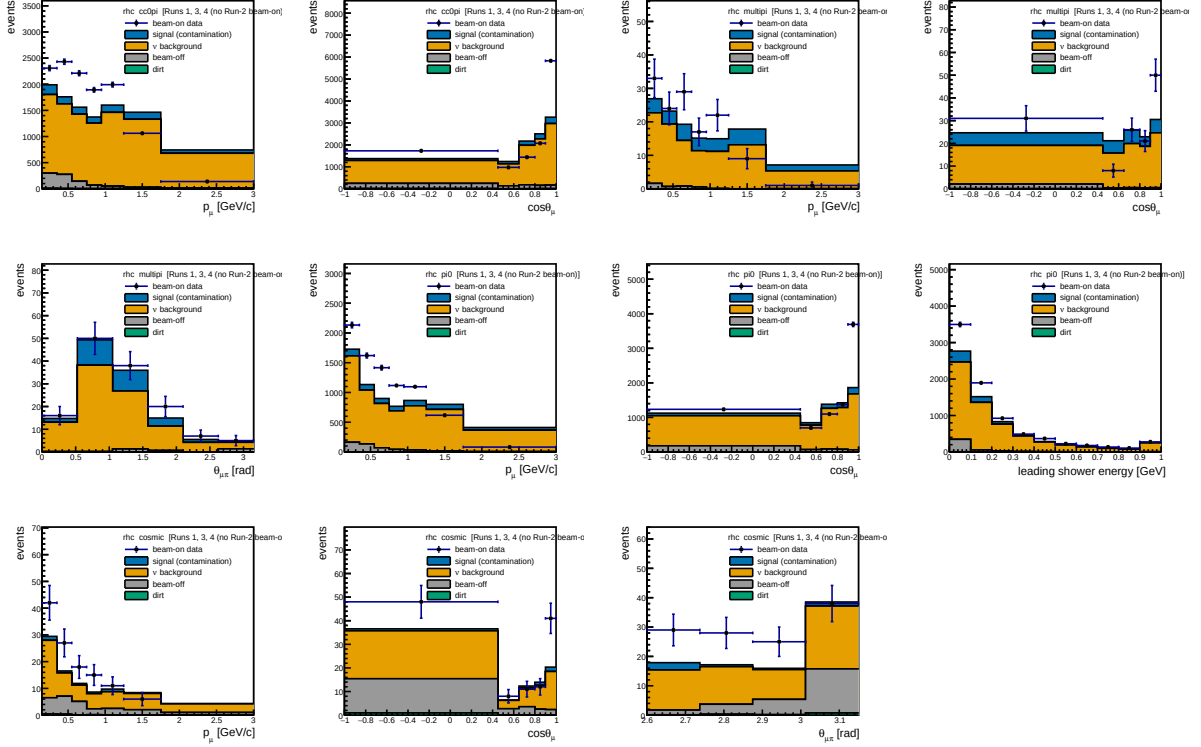


Figure 2: RHC control regions with beam-on data (Runs 1, 3, 4). Same layout as Fig. 1.

3 The run-by-run comparison

The frozen procedure asks for the comparison run period by run period, and it is what makes the mode-level numbers of §2 uninterpretable as they stand.

Table 4: Data over prediction, per run period and region. Each period’s prediction uses that period’s Monte Carlo scaled to its own exposure, its own beam-off sample and gate ratio, and the dirt scaled to its exposure. FHC Run 1 is the outlier; every other period of either mode lies between 0.83 and 1.51, with the high-statistics regions between 1.08 and 1.25.

Period	POT (10^{20})	CC0 π	multi- π	π^0	cosmic
FHC Run 1	3.283	0.637	0.911	0.715	0.595
FHC Run 4	2.075	1.173	1.217	1.248	1.141
FHC Run 5	2.231	1.135	0.899	1.188	1.200
RHC Run 1	0.605	1.191	1.076	1.086	0.828
RHC Run 3	5.003	1.150	1.243	1.246	1.510
RHC Run 4	2.883	1.115	0.823	1.201	1.248

FHC Run 1 is not a systematic of the prediction. The flux uncertainty is a hadron-production model uncertainty common to every run period of a beam mode, and the detector variations are taken from the same Run-4 samples for all periods, so both are almost fully correlated between run periods and cancel in a period-to-period comparison. A prediction systematic can move all FHC periods together; it cannot put Run 1 at 0.64 while Runs 4 and 5 sit at 1.17 and 1.14. The effect is also flat across the four regions of Run 1 — 0.64, 0.91, 0.72, 0.60 — which is the signature of a normalisation applied to the whole period rather than of a mis-modelled background class, since the four regions have very different compositions.

The quantity that is not verified anywhere in this comparison is the Run-1 FHC beam-on exposure itself. The prediction is scaled to 3.283×10^{20} POT and the beam-off to 9846635 beam-on gates, both taken from the trigger table; the beam-on ntuples carry no exposure record of their own, so nothing here cross-checks them against the file that was actually staged. The staged FHC Run-1 file spans runs 4952–6748 and the RHC Run-1 file begins at run 6748, so Run 1 is split between the horn modes inside the run range, which is exactly the situation in which a partial file or a mis-assigned split is easy to miss. The size of the effect is a concrete, falsifiable prediction. Every component of the Run-1 prediction — the Monte Carlo through its POT scale, the beam-off through its gate ratio, the dirt through its exposure — is proportional to the assumed Run-1 exposure, so an exposure error rescales the whole prediction. Bringing FHC Run 1 onto the 1.15–1.20 of the other periods requires the true exposure to be a fraction $f \simeq 0.51$ – 0.62 of the assumed one, that is roughly 1.7 – 2.0×10^{20} POT in place of 3.283×10^{20} , with the three high-statistics regions giving $f = 0.55$ (CC0 π), 0.62 (π^0) and 0.51 (cosmic). The consistency of f across regions of different composition is itself the evidence that a single multiplicative exposure factor, and not a background class, is what is wrong. If instead the trigger table is confirmed against the staged file, the deficit is real and must be treated as a localised reconstruction failure under the response tree.

The remaining five periods carry a common offset. Setting FHC Run 1 aside, the five other periods sit above the prediction by 10–25% in the high-statistics regions, in both beam modes. This is a common normalisation offset. It is within the 20–30% prediction uncertainty, and it is the one direction the global test of §4 is built to be insensitive to, because the correlated flux, cross-section and detector terms absorb it. Two features are worth recording for whatever investigation follows: the offset appears in regions whose beam-off fraction ranges from 3–5% (multi- π) to 28–39% (cosmic), which argues against a beam-off scaling error; and it appears in both horn modes, which argues against a mode-specific flux problem.

4 The frozen comparison procedure applied to the data

The frozen procedure defines one pass/fail statistic — the global normalisation χ^2 over the eight (region, mode) totals with the full pre-fit covariance — and two diagnostics, the per-region pulls and the normalisation-marginalised shape tests of the frozen distributions. All three are computed here with exactly the machinery that was exercised on pseudo-data before opening, with only the data, the retained run periods and the removal of the pseudo-data beam-off throw changed. On pseudo-data that machinery returned a global χ^2 of 0.66/8 and all twenty-two shape tests with $p > 0.47$, so it is not predisposed to fail.

Table 5: Per-region normalisation pulls and the global test. “full” is the pre-fit prediction covariance plus the data Poisson term; “st+det” is the statistical and detector terms alone. The last column is the fractional shift of that region’s total which, on its own, would bring the global χ^2 to its 95% point.

Mode	Region	Data	Prediction	EXT	pull (full)	pull (st+det)	σ_{full}/N
FHC	CC0 π	10 660	11 571.0	9.7%	−0.29	−0.40	27.0%
	multi- π	109	109.6	5.2%	−0.02	−0.03	34.2%
	π^0	6 291	6 327.3	10.0%	−0.02	−0.03	29.6%
	cosmic	92	108.3	38.7%	−0.67	−0.80	22.4%
RHC	CC0 π	12 047	10 558.9	8.0%	0.73	1.75	19.3%
	multi- π	136	125.0	2.9%	0.26	0.41	33.1%
	π^0	8 092	6 639.5	6.9%	0.99	3.14	22.1%
	cosmic	120	89.5	28.2%	1.63	2.21	20.9%
Global normalisation $\chi^2 = 8.77/8$, $p = 0.362$ — passes ($p > 0.05$ required)							
Blind spot: a common shift of all eight totals fails the test only beyond 61.9%							

Normalisation: the set passes. The global χ^2 is 8.77 on eight degrees of freedom, $p = 0.362$. No region is flagged by the $|\text{pull}| > 2$ criterion against the full covariance; the largest is the RHC cosmic region at 1.63. That is the expected outcome given §3: the offsets are 10–25% against a prediction uncertainty of 19–34% per region, and they are strongly correlated between regions (0.4–0.9), which is exactly the direction the test cannot resolve. Against the statistical and detector terms alone, which is the diagnostic column, RHC π^0 reaches 3.14 and RHC cosmic 2.21.

Table 6: Normalisation-marginalised shape tests of the frozen distributions. The overall normalisation of each distribution is projected out before the χ^2 is formed, so these are insensitive to the offsets of §3. Entries with $p < 0.01$ in bold.

Mode	Distribution	χ^2/ndf	p	Mode	Distribution	χ^2/ndf	p
FHC	CC0 π p_μ	13.09/6	0.042	RHC	CC0 π p_μ	52.85/6	0.000
	CC0 π $\cos \theta_\mu$	350.51/4	0.000		CC0 π $\cos \theta_\mu$	583.58/4	0.000
	multi- π p_μ	6.29/6	0.391		multi- π p_μ	10.67/6	0.099
	multi- π $\cos \theta_\mu$	8.49/4	0.075		multi- π $\cos \theta_\mu$	9.97/4	0.041
	multi- π $\theta_{\mu\pi}$	0.75/5	0.980		multi- π $\theta_{\mu\pi}$	0.64/5	0.986
	π^0 p_μ	30.41/6	0.000		π^0 p_μ	67.40/6	0.000
	π^0 $\cos \theta_\mu$	210.36/4	0.000		π^0 $\cos \theta_\mu$	240.85/4	0.000
	π^0 E_{shr}	7.14/9	0.623		π^0 E_{shr}	7.20/9	0.616
	cosmic p_μ	4.54/6	0.604		cosmic p_μ	4.22/6	0.647
	cosmic $\cos \theta_\mu$	5.56/4	0.234		cosmic $\cos \theta_\mu$	4.27/4	0.371
	cosmic $\theta_{\mu\pi}$	2.80/3	0.423		cosmic $\theta_{\mu\pi}$	2.12/3	0.547

Shape: the muon angle fails, and it is not the normalisation offset. The shape tests project the overall normalisation out, so they are blind to everything §3 is about. Six of the twenty-two fail at $p < 0.01$, and they are not scattered: they are the muon momentum and, far more strongly, the muon angle in the two high-statistics regions, CC0 π and π^0 , in *both* beam modes. The $\cos \theta_\mu$ values, 351/4 and 210/4 in FHC and 584/4 and 241/4 in RHC, are not marginal. The remaining sixteen tests, including every distribution of the multi- π and cosmic regions and the π^0 shower energy that the region was built to test, are unremarkable.

Four things bound how far this can be taken.

- **It is not an artefact of the test.** The same code on pseudo-data returned all twenty-two with $p > 0.47$. The failures appear only when real data is substituted.

- **It is not the FHC Run-1 exposure.** A wrong weight on one run period does distort a summed distribution, because the periods do not have identical shapes, so the FHC numbers are not clean. But the largest failures are in RHC, which contains no period suspected of an exposure error.
- **The test is very sensitive by construction.** With 10^4 events per region and a covariance dominated by normalisation-like flux and detector terms, projecting the normalisation out leaves a small shape covariance, so a discrepancy of a few per cent per bin produces a large χ^2 . The size of these numbers should not be read as the size of the effect.
- **The binning concentrates the forward peak.** The released $\cos \theta_\mu$ edges are $-1, 0.45, 0.65, 0.8, 0.9, 1$: the first bin spans three quarters of the range, so the test is largely comparing the forward peak with everything else. A forward-angle discrepancy is where the MicroBooNE BNB CC1 $\pi^\pm$ measurement also found every tested generator above the data.

Procedural consequence. Under the frozen procedure a fitted normalisation correction is permitted only for a region whose shape test passes, so that the discrepancy is consistent with a pure normalisation shift. On these numbers neither CC0 π nor π^0 qualifies: both fail their muon-angle shape test in both modes. Whatever the eventual explanation, a fitted correction to either region is not available under the rules as frozen, and the response tree points instead at the added-uncertainty branch or at a model revision with renewed approval. That conclusion should be revisited once the exposure questions of §3 are closed and the tests are repeated.

5 What this establishes, and what it does not

- The processing chain runs on real beam-on data of both productions, pre-CRT and CRT-era, without error, and the blinding mechanism behaves as designed: no signal-region event was written, and the outputs are refused by the extraction.
- The control-region distributions are complete and are shown for the first time against data.
- **No conclusion about the background model should be drawn yet.** Two bookkeeping issues sit in front of any such conclusion: the missing Run-2 exposure, which is excluded consistently here but leaves 19% of the analysis unexamined, and the FHC Run-1 anomaly, which has the size and the flatness of an exposure error.
- The frozen normalisation test passes. The muon-angle shape test does not, in the two high-statistics regions and in both beam modes; the failure survives the removal of the normalisation, is absent from the same code run on pseudo-data, and is largest in RHC, which carries no suspected exposure error. It is the one result here that the bookkeeping issues do not obviously explain, and it is the thing to understand next after them. Its size should not be read from the χ^2 : the shape covariance is small once the normalisation is projected out, so a few per cent per bin goes a long way, and the released binning puts three quarters of the $\cos \theta_\mu$ range in one bin.
- The frozen procedure's own response tree treats a single-period excursion as a reconstruction or exposure issue to be investigated, not as a model correction, and forbids excluding a period without independent evidence of a defect. That is the path FHC Run 1 should follow.

Required next.

1. Confirm the exposure and gate count of the staged FHC Run-1 beam-on file against the trigger table, and confirm that the FHC/RHC split of Run 1 at run 6748 is the one the exposure table assumes.
2. Establish whether Run-2 beam-on data exist upstream. If they do, stage and process them; if they do not, Run 2 must be removed from the exposure and the whole normalisation regenerated.
3. Repeat this comparison once both are settled, and only then read the normalisation offset of §3 as a statement about the background.
4. Investigate the muon-angle shape failure independently of the exposure work: it is present in RHC alone, so it can be studied on the RHC periods while the FHC Run-1 question is open. A finer $\cos \theta_\mu$ binning of the control regions, which is not the released signal-region binning and is free to be chosen, would show whether the discrepancy is confined to the forward peak.
5. The prerequisites for opening the signal region are unchanged and none of them is met by this note: the alternative-generator closure, the flux central-value confirmation, the MCS momentum bound, the clean-cache reproduction, and the inverted-PID regions for the misidentification transfers.

Provenance. Beam-on skim written 2026-09-09 from the eight files of Table 1; prediction from the release of 2026-09-06 (the same Monte Carlo, beam-off and dirt samples, and the same per-run exposure scaling, as the blind studies), with Run 2 removed from both sides.